Max J. Emerick

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Summary

Ph.D. Candidate in Mechanical Engineering at UC Santa Barbara and NDSEG Graduate Fellow. My research focuses on developing mathematical tools for the modeling, analysis, and control of continuum models of large-scale systems, drawing from the areas of partial differential equations, optimal control, optimal transport, and geometric/variational methods. I have applied these tools across several domains, including swarm robotics, fluid mixing, and coupled-oscillator networks. Currently seeking tenure-track faculty and postdoctoral research positions in dynamical systems and control or applied mathematics.

Education

Ph.D. in Mechanical Engineering, University of California, Santa Barbara — Exp. Sep 2027
Focus in **Dynamical Systems and Control Theory** — **4.0 GPA**
Advisor: Bassam Bamieh

M.S. in Mechanical Engineering, University of California, Santa Barbara — Sep 2022
Focus in **Dynamical Systems and Control Theory** — **4.0 GPA**
Advisor: Bassam Bamieh
Thesis: Control of Continuum Swarm Systems via Optimal Control and Optimal Transport Theory

B.S. in Mechanical Engineering, California Polytechnic State University — Jun 2020
Concentration in **Mechatronics**, Minor in **Mathematics** — **3.9 GPA**
Graduated **Summa Cum Laude**
Capstone Project: Surface Autonomous Vehicle for Emergency Rescue

Honors & Awards

National Defense Science and Engineering Graduate (NDSEG) Fellowship — Sep 2024 – Present

Outstanding TA Award, UCSB Mechanical Engineering Department (Student-Nominated) — May 2025

Outstanding Student Paper Award, 62^{nd} IEEE Conference on Decision and Control — Dec 2023

Best TA Award, UCSB Mechanical Engineering Department (Faculty-Nominated) — Oct 2023

IFAC Young Author Award, 9^{th} IFAC Conference on Networked Systems — Jul 2022

Research Experience

Graduate Student Researcher, Bamieh Lab, UC Santa Barbara — Apr 2021 – Present

- Conducting fundamental research on control of continuum models for large-scale systems using tools from PDEs, optimal control, optimal transport, and geometric/variational methods
- Developing analysis and control techniques for multi-agent, ensemble, and fluid systems
- Acting as primary mentor for two undergraduate students

Student Researcher, Flight Test Data System Lab, Cal Poly — Jun 2019 – Sep 2020

- Designed, programmed, and tested embedded systems for taking real-time measurements of boundary layers on aircraft
- Assisted in writing and maintaining C/C++ libraries for numerous devices and applications
- Designed experiments and performed data analysis for device testing and calibration
- Completed three projects, authored four internal reports, and devised new firmware architecture to accelerate the research group’s progress

Publications

[J4] **"Incompressible Fluid Mixing as Constrained Optimal Transport,"**
M. Emerick, J. Igraszek, and B. Bamieh. In preparation for submission to Journal of Nonlinear Science.

[J3] **"Mean-Field Oscillator Ising Machines: Gradient Flows and Classification of Limit Solutions,"**
B. Bamieh, F. Bullo, **M. Emerick,** and A. R. Venkatakrishnan. In preparation for submission to SIAM Review (available on arXiv).

[J2] **"On Distributed Control of Continuum Swarms: Local Controllers as Differential Operators,"**
M. Emerick, S. P. Chhatoi, and B. Bamieh. In preparation for submission to IEEE Transactions on Automatic Control (available on arXiv).

[J1] **"Optimal Assignment and Motion Control in Two-Class Continuum Swarms,"**
M. Emerick, S. Patterson, and B. Bamieh, IEEE Transactions on Control of Network Systems, vol. 13, no. 1, pp. 130-142, 2026.

[C4] **"Incompressible Optimal Transport and Applications in Fluid Mixing,"**
M. Emerick and B. Bamieh, 64th IEEE Conference on Decision and Control, pp. 3157-3162, 2025.

[C3] **"Causal Tracking of Distributions in Wasserstein Space: A Model Predictive Control Scheme,"**
M. Emerick, J. Jonas, and B. Bamieh, 63rd IEEE Conference on Decision and Control, pp. 7606-7611, 2024.

[C2] **"Continuum Swarm Tracking Control: A Geometric Perspective in Wasserstein Space,"**
M. Emerick and B. Bamieh, 62nd IEEE Conference on Decision and Control, pp. 1367-1374, 2023.

[C1] **"Optimal Combined Motion and Assignments with Continuum Models,"**
M. Emerick, S. Patterson, and B. Bamieh, *IFAC-PapersOnLine*, vol. 55, no. 13, pp. 121-126, 2022.

Conference Presentations

"Wasserstein Gradient Flow Structure in Mean-Field Oscillator Ising Machines" Aug 2026
Invited Talk, 27th Intl. Symp. On Mathematical Theory of Networks and Systems Waterloo, Canada

"Incompressible Optimal Transport and Applications in Fluid Mixing" Dec 2025
Invited Talk, 64th IEEE Conference on Decision and Control Rio de Janeiro, Brazil

"Tracking Control in The Wasserstein Space" Jul 2025
Contributed Talk, 2025 SIAM Conference on Control and Its Applications Montreal, Canada

"Causal Tracking of Distributions in Wasserstein Space: A Model Predictive Control Scheme"
Contributed Talk, 63rd IEEE Conference on Decision and Control Dec 2024, Milan, Italy

"Tracking Control in The Wasserstein Space" Nov 2024
Contributed Talk, 44th Southern California Control Workshop Los Angeles, United States

"Optimal Control of Distributions in Wasserstein Space" Aug 2024
Invited Talk, 26th Intl. Symp. on Mathematical Theory of Networks and Systems Cambridge, UK

"Continuum Swarm Tracking Control: A Geometric Perspective in Wasserstein Space" Dec 2023
Contributed Talk, 62nd IEEE Conference on Decision and Control Singapore

"Optimal Combined Motion and Assignments with Continuum Models" Jul 2022
Contributed Talk, 9th IFAC Conference on Networked Systems Zurich, Switzerland

Teaching Experience

Graduate Teaching Associate, UC Santa Barbara Sep 2024 – Dec 2024

- Developed and taught (as instructor of record) a new TA training course for the UCSB Mechanical Engineering Department

Lead Teaching Assistant (TA), Mech. Eng. Dept., UC Santa Barbara Apr 2023 – Jun 2025

- Selected to serve as Lead TA for the Mechanical Engineering Department, mentoring incoming graduate students and serving as point-of-contact for the department's 70+ TA's
- Developed new TA trainings, mentorship programs, and courses within the department

Graduate Teaching Assistant, UC Santa Barbara Jan 2021 – Jun 2024

- Provided teaching, tutoring, grading support, and hands-on training to students in a variety of courses, including Dynamic Systems Modeling, Vibrations, Controls, Robotics, Mechatronics, Nonlinear Phenomena, and Design for Test Automation

Teaching Assistant, California Polytechnic State University Jan 2017 – Mar 2017

- Provided teaching and tutoring support to three sections of Electricity & Magnetism

Industry Experience

Engineering Intern, Tesla, Fremont, CA Jun 2021 – Sep 2021

- Worked in manufacturing for the castings team on the pilot line for the "Giga Press", the largest aluminum die-casting machine in the world
- Improved process and workflow throughout the casting and finishing lines by adding new stations, improving interfaces, and automating and optimizing processes

Engineering Intern, Panasonic Avionics Corporation, Bothell, WA Jul 2017 – Aug 2017

- Designed and implemented mounting systems for avionics onboard Boeing 747s and Airbus A330s
- Provided on-site engineering support for system installations and retrofits

Professional Affiliations

Engineer-in-Training, State of California (#172585) Dec 2020 – Present

Affiliate Member, International Federation of Automatic Control (IFAC) Aug 2022 – Present

Graduate Student Member, IEEE & IEEE Control Systems Society (CSS) Jul 2022 – Present

- Member, CSS Technical Committee on Distributed-Parameter Systems

Graduate Student Member, Society for Industrial and Applied Mathematics Jul 2022 – Present

- Member, Activity Group on Controls and Systems Theory
- Member, UC Santa Barbara Student Chapter

Service

Graduate Fellowship Writing Consultant, UC Santa Barbara Aug 2025 – Nov 2025

- Provided writing support to graduate students applying to NSF GRFP and NDSEG fellowships

Founding Board Member & Secretary, UCSB Mech. Eng. Grad. Student Assn. Nov 2023 – Jun 2025

- Helped to organize and set up a new departmental graduate student association

- Took meeting notes, kept records, met with department administration, and helped organize events

STEM Outreach Speaker, Various Nov 2016 – Aug 2022

- Helped organize and deliver STEM workshops at various K-12 schools in California and Arizona